

Assignment 12

1. Consider a monopropellant rocket designed to generate a thrust of 100000 N for 30 s. The specific impulse is 200 s, and the chamber pressure is 3 MPa. The specific gravity of the monopropellant is 1.008. A pressurized gas system (at initial pressure of 10 MPa and initial temperature of 300 K) with helium (molar mass 4, specific heat ratio 1.67) is used for the propellant feed. What is the minimum volume of the gas tank required for the adiabatic expansion of the HPG? What is the corresponding mass of the pressuring gas? {Ans.: 1.086 m³, 17.42 kg. To calculate these values, you first must find the volume of the propellant expelled. This comes to 1.517 m³.}
2. A rocket turbine is driven by gases having specific heat ratio of 1.3 and specific heat of 1070 J/kg-K. The turbine entry stagnation conditions are 810 K and 2 MPa. If the discharge pressure from the turbine is 0.14 MPa, calculate its specific work. $\eta_{ts} = 0.70$. {Ans.: 2.78 (10⁵) J/kg}
3. Reconsider Problem 1. If, instead of the gas pressure feed system, a turbo-pump of efficiency 65% is used as the propellant feed, calculate the power required to drive the pump. Assume that the monopropellant tank is maintained at a pressure of 1.2 atm. {Ans.: 0.22 MW}
4. A rocket propellant is stored in a thermally insulated tank at 207 kPa and 85 K. From here, it is supplied to a turbopump. The liquid propellant has specific gravity of 1.5 and has vapor pressure of 101.3 kPa at 85 K. Given that the propellant enters the turbopump with a speed of 8 m/s, check if cavitation is likely to be a problem. The feed lines are well insulated, and the pump inlet is 1 m below the level of the propellant in the tank. The pipe loss is 0.5 m.